INCH-POUND

MIL-B-43154K
INTERIM AMENDMENT 2 (GL)
19 September 1990
SUPERSEDING
AMENDMENT 1
15 September 1989

MILITARY SPECIFICATION

BOOTS, HOT WEATHER

This interim amendment is approved for use within the Department of the Army, with MIL-B-43154K, dated 16 May 1988.

PAGE 1

* 1.1: Delete in its entirety and substitute:

"1.1 Scope. This specification covers two types of mildew resistant combat boots that have a leather and nylon duck upper and a direct molded sole. The type I boot contains a built-in spike resistant insole, and the type II boot has a insole which is resistant to heat transfer."

* 1.2: Delete the first and second sentence and substitute: "The boots shall be of the following types, sizes, and widths as specified (see 6.2). The schedule of sizes is as listed in table I.

Type I - Black with spike-resistant insole
Type II - Tan with heat transference resistant insole"

PAGE 2

2.1.1, under "SPECIFICATIONS MILITARY": Add "MIL-L-10867 - Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored" and "MIL-C-43734 - Cloth, Duck, Textured Nylon".

Delete "MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys".

PAGE 3

* 2.1.1, under "STANDARDS FEDERAL": Add "FED-STD-595 - Colors Used in Government Procurement".

2.1.2, lines 9 to 11: Delete, and substitute the following:

"2-1-1635 - Speed Lace Assembly"

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(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)"

PAGES 3 AND 4

* 2.2, end of paragraph, add:

"COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

(Color cards may be available from the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity)."

PAGE 4

* 2.2, after "D 2240" add:

"D 3574 - Standard Methods of Testing Flexible Cellular Materials - Slab, Bonded, and Molded Urethane Foams"

* Add the following paragraph:

"3.1.1 Standard sample. The standard sample for color sueded flesh appearance shall be a swatch of cattlehide upper (see 6.4)."

* 3.3.1.1, line 1, after "leather.": Delete the remainder of the paragraph.

* Add the following new paragraphs:

"3.3.1.1.1 Type I. The leather for the vamps, counterpockets, quarter front stays, and outside eyelet stays shall be smooth plated, full grain, or corrected grain side leather conforming to treatment A of MIL-L-3122, except that the thickness shall be as specified for the cut parts in table II. The break for the vamp shall be not more than a break pattern of 4 when examined as specified in 4.4.3. The color shall be black. All finishes applied by the boot manufacturer and tanner shall be compatible with one another to ensure a long-lasting low-gloss finish.

3.3.1.1.2 Type II. The leather for the vamps, counterpockets, quarter front stays, and outside eyelet stays shall conform to MIL-L-3122, treatment C, except that:

a. The flesh surface shall be sueded so as to produce a fine uniform nap (see 3.1.1).

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b. The grain surface shall be lightly buffed to remove the surface of the grain only (see 3.1.1).

c. The color shall be desert tan and shall be obtained by drum dyeing (see 3.1.1). The dye shall penetrate into the leather from both the flesh and grain of the total thickness. The finish requirements shall not apply.

d. Shank and pocket areas, along with any other location on the cattlehide side which have loose flesh or raggy fibers shall not be used.

e. The grain side of the leather shall be in contact with the test pad when tested for staining.

f. The minimum chloroform extractables shall be 4 percent.

g. The tear strength requirements for the outside eyelet stays shall not apply.

h. At least 80 percent of the specimens tested shall have stitch tearing strength values of not less than 30 pounds when tested as specified in accordance with MIL-L-10867."

PAGE 5

Table II, after "Eyelet stays" add: "(Type I)". End of table under "Part" and "Thickness (ounce) Minimum, Maximum" add respectively: "Outside eyelet stays (Type II)", "3", and "4".

* 3.3.2.1: Delete and substitute:

"3.3.2.1 Nylon duck. The cloth for the quarters and gussets shall be nylon duck conforming to class 1 of MIL-C-43734, except that the color shall be Black 375 for Type I and Tan 495 for Type II and the air permeability requirement shall not apply. The nylon duck cloth may be treated with a non-fray, non-stiffening finish to facilitate cutting and handling."

* 3.3.2.2: Delete and substitute:

"3.3.2.2 Backseam tape. The tape shall conform to Type I, class 3, 1/2 or 17/32 inch wide of DDD-T-86. The color shall be Black 375 for Type I and Tan 495 for Type II."

3.3.2.3: Delete and substitute:

"3.3.2.3 Box toes. The box toe material shall be a laminate of 0.020 ± 0.002 inch virgin ionomer resin (Surlyn) sandwiched between two layers of non-woven polyester saturated with polystyrene. The total thickness shall be 0.056 ± 0.005 inch. The box toes shall be cut to the patterns provided and shall be skived 3/8 ± 1/8 inch wide at breast. Testing shall be as specified in 4.4.1."

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PAGE 6

* 3.3.2.5, line 3: Delete "Olive Green 106" and substitute "Black 375 for Type I and Tan 495 for Type II."

* 3.3.2.6, line 5: Delete "Olive Green 106" and substitute "Black 375 for Type I and Tan 495 for Type II."

3.3.3.1, under "Steel shank": Delete the second sentence and substitute: "The width of the shank shall be 1 ± 1/32 inch, or as an alternate for boot sizes 3 - 6-1/2 all widths, the shank width may be 5/8 ± 1/32 inch."

PAGE 7

* 3.3.3.3: Delete and substitute:

"3.3.3.3 Eyelets. The eyelets shall be gloss finish nylon coated brass conforming to commercial size AA. The eyelets shall be black for Type I and Tan 20318 of FED-STD-595 for the Type II."

3.3.3.3.1: Delete.

* 3.3.3.4, end of paragraph title add: "(Type I only)"

PAGE 8

Add the following new paragraph:

"3.3.3.4.2.2 Finish of drainage eyelets. The drainage eyelets for the Type I boot shall have a black oxide finish."

* 3.3.3.5, end of paragraph title add: "(Type I only)"

* Add new paragraph:

"3.3.3.5.1 Thermal barrier. The thermal barrier for the Type II boot shall be 1/8 inch thick, plus 1/32 inch polyurethane foam having a density of 22 ± 7 pounds per cubic feet, a tensile strength of 115 pounds per square inch minimum, and maximum compression set of 22 percent when tested as specified in Table VIII."

Add the following new paragraph:

"3.3.3.6 Speed lace assembly. The speed lace assembly shall be of two-piece construction consisting of gloss finish nylon coated brass speed lacer and nylon coated alternate full tubular rivet in accordance with Drawing 2-1-1635."

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* 3.3.4, subparagraph a., delete in its entirety and substitute:

"a. The color shall be black for Type I and tan for the Type II boot. The compounds shall be nonmarking, producing only a slight erasable mark on white paper."

PAGE 9

3.3.5.1, line 4: Add "and 308" after "301".

* Delete the last sentence and substitute "The color of the thread shall be black for Type I and tan C.A. 66019 for Type II."

3.3.6.1, Table IV, opposite "Thickness", under columns "A" through "D": Delete "9/16" and substitute "19/32".

PAGE 10

* 3.3.6.3, line 2, after "figure 1": Add "and figure 4 for Type I and figure 4 and figure 5 for Type II.".

* Line 3 after "tags": Add "for the Type I boot only".

* 3.3.6.4: Delete and substitute:

"3.3.6.4 Laces. The laces shall conform to type III, class 1 of V-L-61. The color of the laces shall be black for Type I and Tan 495 for Type II. The requirements for colorfastness to water do not apply for the Type II lace. The length of the laces shall be a minimum of 66 inches for all sizes."

Add the following paragraph:

"3.3.6.5.1 Alternate cushion insole (Type II only). As an alternate, the cushion insole shall be a medium brown open celled polyether or polyester polyurethane foam with an integral skin, laminated to a neoprene impregnated cellulose substrate. The combined thickness of the polyurethane foam and cellulose substrate shall be 0.130 ± .013 inches. The polyurethane layer shall have a thickness of 0.105 ± 0.013 inches, and shall have a density of 17 +2, -1 pounds per cubic feet, a minimum tensile strength of 115 pounds per square inch, a maximum 10 percent compression set, and a maximum 5 percent compression set after hydrolysis 1/. The cushion insole shall be one-half size less than the boot in the same width. For size 3, the insole shall conform to the leather insole pattern. The size and width of each insole shall be legibly stamped on the bottom (cellulose substrate side) of the insole in the shank area. The size shall be stamped in numbers and the width in letters. The numbers and letters shall be not less than 3/8 inch in height."

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1/ Less than 5 percent after five hours exposure to saturated steam at 15 pounds per square inch.

3.4: Delete and substitute:

* "3.4 Design.

3.4.1 Type I. The Type I boot shall be a blucher pattern design having a partial speed lace closure and an integral steel sole plate and a removable ventilating insole. The leather areas on the outside of the boot shall have the grain surface exposed. The boot shall have a nylon cloth quarter with nylon webbing for the backstay, top stay, and ankle reinforcements. The vamp shall be plain with a surlyn box toe, and the outersole shall have bars and cleats in accordance with Drawing 2-1-962 (see 2.1.2).

3.4.2 Type II. The design for the type II boot shall be a blucher pattern design having a partial speed lace closure and an integral thermal barrier sole piece and removable ventilating insole. The leather areas on the outside of the boot shall have the flesh surface exposed. The boot shall have a nylon cloth quarter with nylon webbing for the backstay, top stay, and ankle reinforcements. The vamp shall be plain with a surlyn box toe, and the outersole shall have bars and cleats in accordance with Drawing 2-1-962 (see 2.1.2). There will be no drainage eyelets on the vamp."

PAGES 10 and 11

3.5: Delete the fifth sentence.

PAGE 11

Table V, after "Vamp,", under "Component parts": Delete "full toe with nicks for box toe replacement".

* After "steel plate insole" add: "(Type I only)".

PAGE 12

* 3.7.1.1, line 2, end of first sentence: Add "for the Type I boot and flesh out grain leather for the Type II boot."

3.7.1.2: Delete and substitute:

"3.7.1.2 Cutting nylon duck uppers. The nylon duck quarters shall be cut in the warp direction as indicated in figure 2. The nylon duck gusset may be cut in the warp, filling, or bias direction. The nylon duck cloth may be treated with a non-fray, non-stiffening finish to facilitate cutting and handling."

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* Add the following new paragraph:

"3.7.2.1 Finishing leather parts. All exposed leather raw edges shall be stained black for Type I and tan for Type II using materials specified in 3.3.6.6.3."

PAGE 13

3.7.4, line 3, delete "outside" and substitute "inside".

* Line 5, after "contract, and": Add "for the Type I boot only"

PAGE 14

3.7.5.1: Delete sentences 7 and 8 and substitute: "The nylon duck quarters shall be closed at the back from top to bottom using seam type FSa-1 with a 3/8 ± 1/16 inch wide zig-zag stitch type 308, with 9 to 11 stitches per inch. This seam shall be rubbed down and reinforced with stitched tape on the inside of the quarters covering the zig-zag stitching."

PAGE 15

3.7.5.2, lines 1 and 2: Delete "and cotton".

3.7.6: Delete and substitute:

"3.7.6 Vamping and barring. Vamping shall be done with two rows of stitching (single or double needle) close on the edge of the front edge stay, plus two rows spaced approximately 3/16 inch between inside rows. There shall be two single or double bar rows 5/8 ± 1/16 inch in length at each blucher ear. The bar row shall be placed outside and above vamping by 1/16 to 1/8 inch and parallel to the vamping row at blucher point. This barring may be done by the automatic method with tying or locking ends."

* Add the following new paragraphs:

"3.7.6.1 Inserting drainage outlets (Type I boot only). Two drainage eyelets, as specified in 3.3.3.4.1, shall be inserted through the vamp lining on the inside shank of each boot and shall be located just above the insole on the finished boot. Two washers, as specified in 3.3.3.4.2 shall be used to clinch each drainage eyelet. The drainage eyelet shall be inserted with the flat side (perforated) on the grain side of the leather and the cup side innermost. The hole in the vamp through which the drainage eyelet is inserted shall be no more than 7/32 inch in diameter. As an alternative, two drainage washers, as specified in 3.3.3.4.2.1, shall be inserted through the vamp lining on the inside of each boot and shall be located just above the insole on the finished boot. One drainage eyelet, as specified in 3.3.3.4.1.1, shall be used to clinch

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each drainage washer. The drainage washer shall be inserted in the innermost side of the vamp and the drainage eyelet (perforated) on the grain side of the leather. The hole in the vamp through which the drainage washer is inserted shall be not more than 9/32 inch in diameter.

3.7.6.2 Attaching speed lace assembly and eyeletting. On each quarter, there shall be two nylon coated double A eyelets for all sizes spaced evenly from blucher nose to the first speed lace loop. The edge of each eyelet shall be 3/16 to 5/16 inch from the edge of the quarter. For sizes 3 through 6-1/2, there shall be four speed lace loops equally spaced on each quarter above the eyelets, five speed lace loops for sizes 7 through 11-1/2, and six speed lace loops for sizes 12 through 14. The speed lace loops shall be placed perpendicular to the edge of the quarters and parallel to each other, and shall be securely riveted 1/4 ± 1/32 inch from the center fo the rivet to the edge of the quarter."

PAGE 16

3.7.7: Delete and substitute:

"3.7.7 Lacing for lasting. The machine lacing shall provide for an opening after lasting of 3/4 ± 1/8 inch. The quarter shall be laced in the two lower pairs of eyelets."

* 3.7.8.1, line 1: Delete paragraph title and substitute: "Insole construction assembly (Type I only)."

* Add the following new paragraph:

"3.7.8.2 Insole construction assembly (Type II only). The insoles shall be assembled in three layers. The 4-iron leather insole shall have the grain surface next to the foot. The thermal barrier shall be placed in the middle, and the 3-iron leather insole shall be placed on the bottom."

PAGE 17

3.7.9, line 20: Delete "pyroxylin or polystyrene type".

3.7.10: Add "As an alternate, the use of a heat setting machine may be used to thoroughly dry the counters and uppers. The heat setting time shall be approved by the contracting officer."

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PAGE 20

3.7.17.1.2: Delete and substitute:

"3.7.17.1.2 Biscuit molding conditions. The biscuit molding equipment shall be operated so as to exert a minimum pressure of 250 pounds per square inch (psi) on the compound. The sole mold shall be heated to a temperature not to exceed 350°F, and the side mold shall be heated to a temperature not to exceed 300°F. The dwell time at these temperatures during vulcanizing shall be a minimum of 15 minutes. The boots shall be molded with the use of metal lasts which shall conform to the measurements contained in MIL-L-41815."

* 3.7.22: Delete and substitute:

"3.7.22 Instruction tags.

3.7.22.1 Type I. Instruction tags to be tied to the finished boots shall be printed using the data specified on figures 1 and 4.

3.7.22.2 Type II. Instruction tags to be tied to the finished boots shall be printed using the data specified on figures 4 and 5."

PAGE 26

Table VIII: Opposite "Eyelets": Delete the following characteristics and all information related thereto:

"Outside diameter of flange and barrel",
"Diameter of hole",
"Overall length",
Thickness".

* Table VIII, after "Drainage eyelets 1 pound" add: "(Type I only)".

PAGE 27

* Table VIII, after "Washers" add: "(Type I only)".

PAGE 28

* Table VIII: Add the following in their respective columns:

"Speed lace loops	Material identification	3.3.3.6	1/	------
Thermal barrier	Thickness	3.3.3.5.1	1/	------
	Density	3.3.3.5.1	D 3574 1/ 4/	-1----
	Tensile strength	3.3.3.5.1	D 3574 1/ 4/	-1----
	Compression set	3.3.3.5.1	D 3574 1/ 4/	-1----

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Insole, cushion alternate (1 sq. yd. material)	Material identification	3.3.6.5.1	1/	------
	Thickness	3.3.6.5.1	D 3574 1/ 4/	------
	Density	3.3.6.5.1	D 3574 1/ 4/ (Test A)	------
	Tensile strength	3.3.6.5.1	D 3574 1/ 4/ (Test E)	------
	Compression set	3.3.6.5.1	D 3574 1/ 4/ (Test D)	------
	Compression after hydrolysis	3.3.6.5.1	D 3574 1/ 4/ (Test D)"	------

PAGE 29

* 4.4.1.1, end of title add: "(Type I only)".

32

* 4.4.2.3, after paragraph title, add: "(Type I only)"

PAGE 33

* 4.4.3, fourth sentence, delete "ventilating".

PAGE 34

Table XII, under "Examine" column for "Upper leather", opposite "Defect" for "Fat wrinkles, flanky leather, or veins": Delete "Major" classification and substitute "*".

PAGE 35

Table XII, under "Examine": Delete "Nylon and cotton quarter and gusset" and substitute "Nylon quarter and gusset".

PAGE 36

Table XII, under "Examine" column for "Vamp stitching at outersole junctions", opposite "Defect" for "More than the specified maximum number of stitches": Delete "Major" classification and substitute "*".

PAGE 39

Table XII: Add the following to the indicated columns:

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Examine	Defect	Classification Major	Minor
"Speed laces	Number of speed lace loops not as specified but each two has the same number		X
	Not the same number of loops in each row	X	
	Loops not properly spaced within the row or misaligned between the rows to an extent interfering with proper lacing		*
	Loops not perpendicular to edge of quarters or parallel to each other		X
	Any loop not securely riveted	X	
	Distance from center of rivet to edge of quarter less than 7/32 or more than 9/32 inch		*
	Four splits in rivet clinch roll		X
	Five or more splits in rivet clinch roll	X	
Color and finish	Any raw edges not stained to match upper leather		X"

PAGE 45

* 6.1: Delete and substitute:

"6.1 Intended use. The type I boots are intended for use in hot, wet tropical areas and the type II boots are intended for use in hot, dry desert areas by military personnel of the Department of Defense."

* 6.2.b: Delete and substitute:

"b. Type, size and width required (see 1.2)."

Add the following paragraphs:

"6.7.1 Suggested source for cushion insoles. A suggested source of supply for cushion insoles meeting the requirements of 3.3.6.5 is Rogers Corp., Poron Division, East Woodstock, CT 06244.

6.7.2 Suggested source for thermal barrier. A suggested source of supply for thermal barriers meeting the requirements of 3.3.3.5.1 is Foamex Corp., (Custom Felt 10-900C), Eddystone, PA."

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PAGE 48

* Figure 1, line 1, after "INSTRUCTION TAG": Add "FOR TYPE I".

Instruction 2. Sizes., line 2: Delete "3R and 3-1/2" and substitute "3R and 3-1/2R".

Instruction 3: Delete and substitute:

"3. Pull on boots, seating heel firmly into place, then lace. Boots should fit snugly but not tightly. There should be approximately 1/2 inch additional length at toe."

PAGE 51

Add new figure 4.

PAGE 52

Add new figure 5.

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:

Army - GL

(Project 8430-A398)

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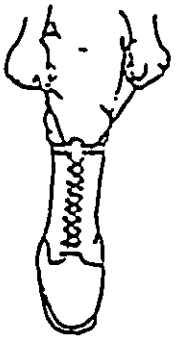
THE FOLLOWING ARE LACING INSTRUCTIONS FOR SPEED LACES



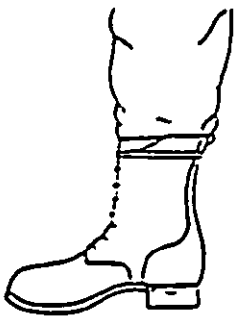
STEP 1
After first lacing through all loops and tying each end of the lace with a hard knot. Loosen up lace for easy step-in of foot.



STEP 2
With thumbs on either edge remove all pleats and wrinkles by pushing together securely around leg.



STEP 3
Hold lace ends at pre-tied knots. Pull up lace to desired tightness.



STEP 4
Wrap extra lace length around collar and secure.

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INSTRUCTION TAG (FOR TYPE II)

BOOTS, HOT WEATHER

INSTRUCTION FOR USE

1. Wear with: 1 Pair socks, wool, cushion sole
1 Pair insoles, ventilating

2. Sizes. The boot is supplied in whole and half sizes 3 through 14 in all widths except sizes 3XN and 3-1/2XN, 3N, and 3-1/2N, 3R and 3-1/2R, 3W and 3-1/2 W, and 4XN and 4-1/2XN. In some instances it may be necessary to make a size adjustment in order to obtain an adequate fit due to normal swelling of the feet and the use of ventilating or cushion insoles.

3. Pull on boots, seating heel firmly into place, then lace. Boots should fit snugly but not tightly. There should be approximately 1/2 inch additional length at toe.

4. Trouser should be worn outside the boot.

5. Carry extra socks with you. Change socks often, at least once daily. Remove boots, and dry feet as often as possible. Neglect of the feet can make you a casualty.

6. Remove ventilating insoles, wash insoles in warm (not boiling) soapy water and shake out to dry.

7. Brush mud and excessive dust off boots. These boots have a sueded surface and are not designed to be polished.

8. Wipe inside of boots occasionally with a moist, soapy rag, and let dry.

9. As frequently as possible, dry boots and insoles in the sun. Do not place near intense heat as this will destroy nylon, leather, and plastic.

10. Before putting on boots in snake, insect, scorpion, or spider infested areas, carefully inspect the inside of the boots to avoid painful and disabling bites and stings.

FIGURE 5. Hot weather boots instruction tag (Type II only)

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